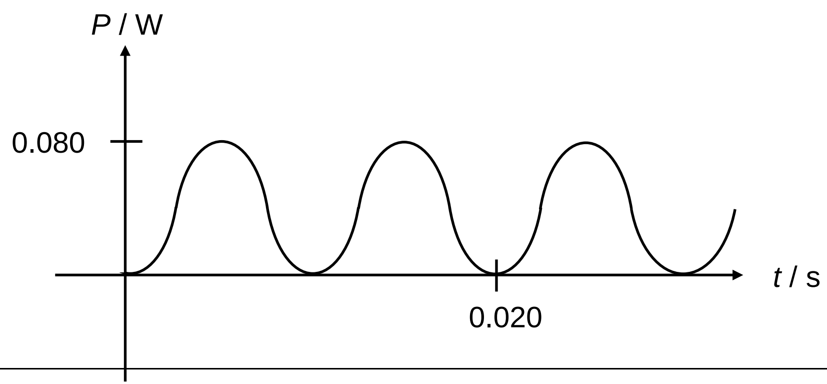
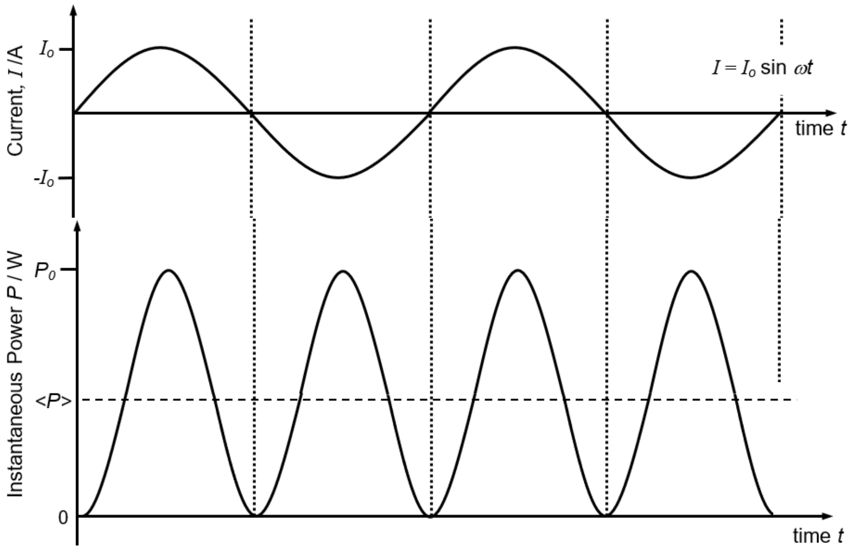
9a	$V_{r.m.s.} = \sqrt{\dfrac{2^2(0.002)+1^2(0.002)}{0.01}}$ $= 1.0$ V	M1 A1
9b	Transmission of electrical energy at high voltage means that the <u>current is low</u> (according to $P = IV$) for the same power. Power <u>loss</u> through joule heating (I^2R) is hence lowered as less electrical energy is dissipated as <u>heat/thermal energy</u> in the cables of resistance R.	M1 A1
9ci	$\dfrac{V_s}{V_p} = \dfrac{N_s}{N_p}$ $V_s = 71 \times 6.5 \times 10^{-3} = 0.46$ V	A1
9cii		

<table>
<tr><td></td><td>Correct shape: sine squared or cosine squared.
Correct labelling of values: period of 0.02 s and peak power of 0.080 W.

<u>Explanation:</u>
Power dissipated in the resistor, $P = I^2R$ (square of a sine function)


The average power of 0.040 W is half that of peak power. Hence, peak power is 0.080 W. Also, the period T = 1/f = 1/50 = 0.020 s.</td><td>B1
B1</td></tr>
<tr><td>9ciii</td><td>The current flow direction alternates between downwards through R and no current flow

with equal time intervals of 0.01 s / for half a period.</td><td>B1

B1</td></tr>
<tr><td>9civ</td><td>Since each sheet is insulated from the others, lamination <u>decreases</u> the size of <u>eddy/induced currents</u>.

With the same induced emf, the <u>heat/thermal energy</u> <u>losses</u> in the iron core will be <u>reduced</u>.</td><td>M1

A1</td></tr>
<tr><td>9di</td><td>An ideal gas is one which obeys the equation of state <u>pV = nRT at all pressures, volumes and temperatures</u>.</td><td>A1</td></tr>
<tr><td>9dii</td><td>
<table>
<tr><th></th><th>work done on gas / J</th><th>heat supplied to gas / J</th><th>increase in internal energy of gas / J</th></tr>
<tr><td>A to B</td><td>+360</td><td>0</td><td>+360 &</td></tr>
<tr><td>B to C</td><td>0 $</td><td>+670</td><td>+670 $</td></tr>
<tr><td>C to D</td><td>−810 &</td><td>0</td><td>−810</td></tr>
<tr><td>D to A</td><td>0 @</td><td>-220 @</td><td>−220 #</td></tr>
</table>

&: first and third line correct
increase in internal energy = work done on gas + heat supplied to gas

$: second line correct
work done on gas (B to C at constant volume) = 0 J
increase in internal energy = 0 + 670 = 670 J</td><td>

B1

B1

B1</td></tr>
</table>

[Turn Over

	#: −220 correct in right hand column total change in internal energy = 0 J (cyclical process) thus increase in internal energy (D to A) = 810 – 360 – 670 = – 220 @: other two figures correct in last line work done on gas (D to A at constant volume) = 0 J thus heat supplied to gas (D to A) = – 220 J	B1
9diii	Consider the process D to A, Since the gas is ideal, $\Delta U = \frac{3}{2}nR\Delta T$ From the ideal gas equation, since V is constant, $V\Delta p = nR\Delta T$ For the process D to A, $\Delta U = \frac{3}{2}V\Delta p$ ➔ $-220 = \frac{3}{2}(750\times10^{-6})(1\times10^{5} - p_D)$ ➔ $p_D = 2.955\times10^{5}$ Pa Using $p_D V_D = nRT_D$ $n = \frac{p_D V_D}{RT_D} = \frac{2.955\times10^{5}\times750\times10^{-6}}{8.31\times888} = 0.030$ mole	C1 A1
9div	the <u>gas molecules bounce off the receding/outward moving piston</u> hence a <u>decrease in kinetic energy /</u> <u>lower speeds</u> of the gas molecules, leading to lower temperature	B1 B1